

Assignment : Time and Space Complexity

Problem 1 :

```
int a = 0, b = 0;
for (i = 0; i < N; i++) {
    a = a + rand();
}
for (j = 0; j < M; j++) {
    b = b + rand();
}
```

Problem 2 :

```
int a = 0;
for (i = 0; i < N; i++) {
    for (j = N; j > i; j--) {
        a = a + i + j;
    }
}
```

Problem 3 :

```
int a = 0, i = N;
while (i > 0) {
    a += i;
    i /= 2;
}
```

Problem 4 :

```
for(var i=0;i<n;i++)
    i*=k
```

Problem 5 :

```
var value = 0;
for(var i=0;i<n;i++)
    for(var j=0;j<i;j++)
        value += 1;
```